	GLOBAL ECONOMICS **Grade: 11th - Learning Evidence C5, C6 and C7** **Continuous distributions and sample data** **Teacher: Nicolás López Cuéllar**	**1st TERM** **2026–2027**

Learning objective: Use continuous probability models and sample data to describe typical values, variability, and interval probabilities, and interpret differences between sample estimates and theoretical values.	**Assessment criteria:** C5: Calculates and interprets the mean, variance and standard deviation of continuous distributions using standard formulas and functions. C6: Estimates interval probabilities using relative frequencies from generated samples. and compares them with theoretical probabilities. C7: Calculates sample means, variances, and standard deviations for different sample sizes and data representations.

Name: _____ **Grade:** _____ **Date:** _____

Criteria assessment

This exam assesses C5, C6, and C7 separately (6 points each; 18 total). Each task specifies its points. Partial credit requires correct procedures; answers without procedures receive zero. Missing names or late submissions incur a 1-point deduction per criterion.**Passing requires at least 4 points per criterion**. Use unrounded intermediate values and round final statistics to 3 decimals and percentage errors to 2.

Question	C5	C6	C7
1	2	–	–
2	2	–	–
3	2	–	–
4	–	2	–
5	–	1	–
6	–	2	–
7	–	1	–
8	–	–	2
9	–	–	1
10	–	–	1
11	–	–	2
Total	**6**	**6**	**6**

1. Two custom-print studios [C5-2]

Two studios print custom posters. Each order takes between 3 and 15 hours. The owners use smooth outlines to describe how often different completion times occur over many orders; times outside these limits never occur. At studio I, longer jobs are more common: the outline is a single straight edge rising from zero at 3 hours to its highest point at 15 hours. At studio D, shorter jobs are more common: the outline is a single straight edge falling from its highest point at 3 hours to zero at 15 hours. Let I and D be the respective completion times in hours. Identify each distribution from the clues and calculate its mean, variance, and standard deviation using the corresponding formulas. **[2]**

2. Bookings and daily takings [C5-2]

A camera repair shop receives online bookings throughout its opening hours, averaging 10 bookings per hour. Bookings arrive independently, one at a time, at a steady pace. In a very short time window, the chance of a booking is proportional to the window's length; two or more bookings are negligible. Past waiting does not affect the chance of a booking during the next minute. Let T be the time between consecutive bookings in minutes.

On comparable working days, total revenue varies because of many small, independent influences, with none dominating. Its smooth outline has a symmetric, centered at \$3200, with a standard deviation of \$200. Let R be daily revenue in hundreds of dollars.

Identify the distributions for T and R from the clues and use their corresponding formulas to calculate the mean, variance, and standard deviation for T in minutes and for revenue in dollars. Interpret the mean revenue. **[2]**

3. Two parcel-labeling stations [C5-2]

A delivery company uses two parcel-labeling stations. At station A, an automated machine labels a parcel in anywhere from 5 to 19 minutes. Within these limits, any two time windows of the same length are equally likely to contain the finishing time. Let A be its labeling time in minutes.

At station B, staff label special parcels, also taking between 5 and 19 minutes. The manager draws a smooth outline of how often different labeling times occur. The outline starts at zero at 5 minutes, rises along a straight edge to its single highest point at 10 minutes, then falls along a straight edge to zero at 19 minutes. No labeling times occur outside these limits. Let B be this labeling time in minutes. Identify the distributions for A and B from the clues and calculate their respective means, variances, and standard deviations using the corresponding formulas. **[2]**

4. File upload times, grouped data [C6-2]

A sample of file upload times in seconds is summarized below. Classes are left-inclusive and right-exclusive.

Class	$[2, 8)$	$[8, 14)$	$[14, 20)$	$[20, 26)$
f_i	5	9	7	3

The theoretical probability is $P(8 \le X < 20) = 0.75$. Find the empirical probability of the stated interval from the given sample and calculate its absolute and percentage errors relative to the given theoretical probability. **[2]**

5. Office electricity use, raw data [C6-1]

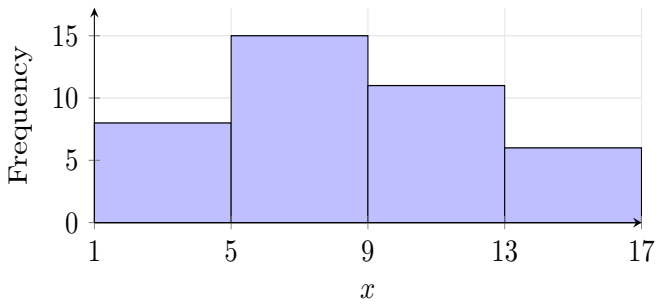
A simulated sample of daily office electricity use (tens of kilowatt-hours) is

1.3	1.7	2.0	2.2	2.6	2.9	3.3	3.5	3.8	4.2
1.6	1.9	2.3	2.7	3.1	3.4	3.6	3.9	4.4	4.6

The theoretical probability is $P(2 \le X \le 3.5) = 0.60$. Find the empirical probability of the stated interval from the given sample and calculate its absolute and percentage errors relative to the given theoretical probability. **[1]**

6. Museum entrance queues, histogram data [C6-2]

The histogram is the only record of a sample of museum entrance queue times. Bar heights are frequencies; times are in minutes, with left-inclusive, right-exclusive classes.



The theoretical probability is $P(5 \le X < 13) = 0.70$. Find the empirical probability of the stated interval from the given sample and calculate its absolute and percentage errors relative to the given theoretical probability. **[2]**

7. Equipment rental times, frequency data [C6-1]

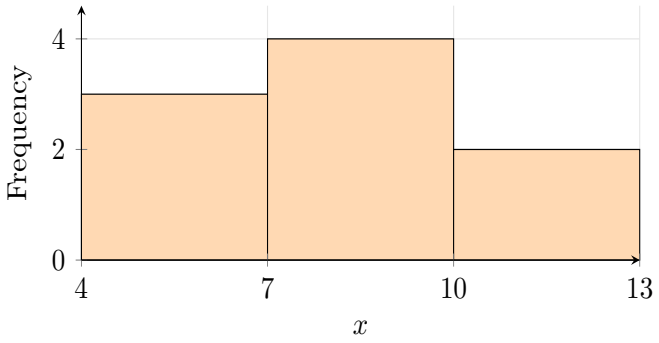
Recorded equipment rental times are summarized below.

x_i (min)	4	8	12	16	20	24
f_i	8	16	28	22	14	12

The theoretical probability is $P(8 \le X < 20) = 0.80$. Find the empirical probability of the stated interval from the given sample and calculate its absolute and percentage errors relative to the given theoretical probability. **[1]**

8. Luggage masses, histogram data [C7-2]

A histogram gives a sample of luggage masses in kilograms. Bar heights are frequencies; classes are left-inclusive and right-exclusive. The model has $\mu = 9$ kg and $\sigma^2 = 7$ kg^2.



Calculate the sample mean, sample variance, and sample standard deviation from the given data, then calculate the absolute and percentage errors of each statistic relative to its theoretical value. **[2]**

9. Chamber temperatures, raw data [C7-1]

Five chamber temperature readings in degrees Celsius are $14, 17, 18, 19, 22$. The theoretical model has $\mu = 19$ degrees Celsius and $\sigma^2 = 6$ square degrees Celsius. Calculate the sample mean, sample variance, and sample standard deviation from the given data, then calculate the absolute and percentage errors of each statistic relative to its theoretical value. **[1]**

10. Parcels dispatched, frequency data [C7-1]

The numbers of parcels dispatched in nine short collection periods are summarized below. The theoretical model has $\mu = 7$ parcels and $\sigma^2 = 9$ parcels^2.

x_i	3	7	11
f_i	2	4	3

Calculate the sample mean, sample variance, and sample standard deviation from the given data, then calculate the absolute and percentage errors of each statistic relative to its theoretical value. **[1]**

11. Walking times, grouped data [C7-2]

Walking times in minutes are grouped using left-inclusive, right-exclusive classes. The theoretical mean is 23 minutes and variance is 30 minutes^2.

Time (min)	$[12, 18)$	$[18, 24)$	$[24, 30)$
f_i	2	3	4

Calculate the sample mean, sample variance, and sample standard deviation from the given data, then calculate the absolute and percentage errors of each statistic relative to its theoretical value. **[2]**